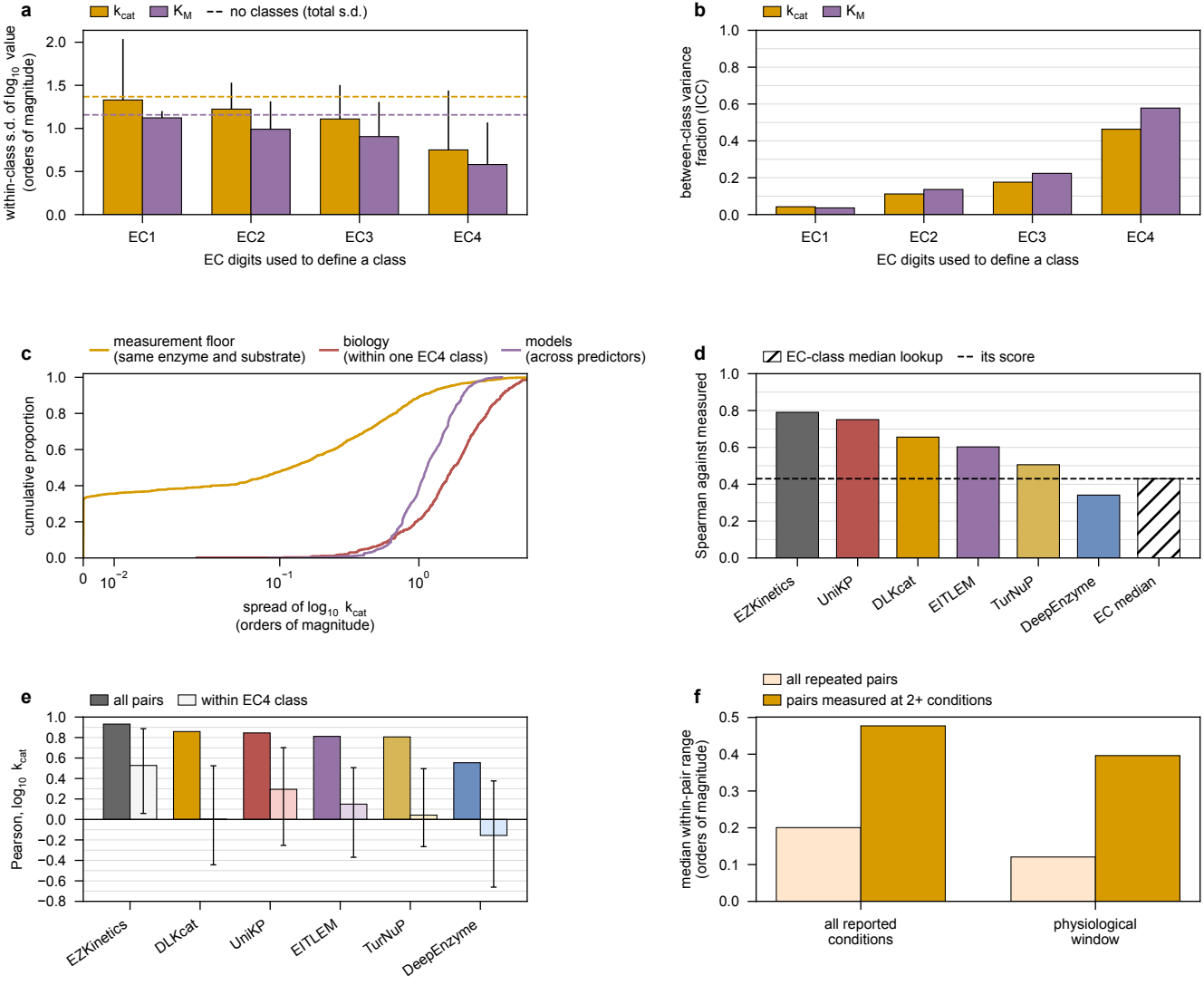
a
k_{cat}
K_M
no classes (total s.d.)
within-class s.d. of $\log_{10}$ value (orders of magnitude)
2.0
1.5
1.0
0.5
0.0
EC1
EC2
EC3
EC4
EC digits used to define a class
b
k_{cat}
K_M
between-class variance fraction (ICC)
1.0
0.8
0.6
0.4
0.2
0.0
EC1
EC2
EC3
EC4
EC digits used to define a class
c
measurement floor (same enzyme and substrate)
biology (within one EC4 class)
models (across predictors)
cumulative proportion
1.0
0.8
0.6
0.4
0.2
0.0
0
10^{-2}
10^{-1}
10^{0}
spread of $\log_{10}$ k_{cat} (orders of magnitude)
d
EC-class median lookup
its score
Spearman against measured
1.0
0.8
0.6
0.4
0.2
0.0
EZKinetics
UniKP
DLKcat
EITLEM
TurNuP
DeepEnzyme
EC median
e
all pairs
within EC4 class
Pearson, $\log_{10}$ k_{cat}
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
EZKinetics
DLKcat
UniKP
EITLEM
TurNuP
DeepEnzyme
f
all repeated pairs
pairs measured at 2+ conditions
median within-pair range (orders of magnitude)
0.5
0.4
0.3
0.2
0.1
0.0
all reported conditions
physiological window